

Sharp sampling numbers for nonseparable reproducing-kernel Hilbert spaces

Automated Math Discovery Run `fa_banach_001`; Model: GPT5.6
9 August 2026

Status: full proof candidate. Remark 13 of Dolbeault–Krieg–Ullrich predicts that separability can be removed from their sharp L_2 sampling theorem by adding $\text{tr}_0(K)/m$. The theorem below proves exactly that prediction. The proof is nonconstructive and the displayed universal constant is intentionally not optimized.

1. Source question and statement

Let $(D, \mathcal{A}, \mu)$ be a measure space and let $H = H(K)$ be a (not necessarily separable) RKHS of measurable complex-valued functions on D , with K measurable on $(D \times D, \mathcal{A} \otimes \mathcal{A})$. Assume

$$\text{tr}(K) := \int_D K(x, x) d\mu(x) < \infty.$$

Then the canonical map $T : H \rightarrow L_2(\mu)$ is Hilbert–Schmidt. Write its positive singular values, with multiplicity, as $\sigma_0 \geq \sigma_1 \geq \dots > 0$, and put $\sigma_k = 0$ after the end when the rank is finite. For the unit ball $F = B_H$, use the source convention $d_k(F) = \sigma_k$ for its L_2 Kolmogorov widths.

Let $H_0 = \ker T$, let K_0 be its reproducing kernel, and set

$$\tau_0 := \text{tr}_0(K) := \int_D K_0(x, x) d\mu(x) = \text{tr}(K) - \sum_{k \geq 0} \sigma_k^2 \geq 0. \quad (1)$$

The sampling number $g_n(F)$ is the least worst-case L_2 error of a linear algorithm using at most n point values.

Theorem 1 (nonseparable sharp sampling bound). *There is a universal integer C such that, for every $n \geq 1$,*

$$g_{Cn}(B_H)^2 \leq \frac{1}{n} \left(\sum_{k \geq n} d_k(B_H)^2 + \text{tr}_0(K) \right). \quad (2)$$

One admissible value furnished by the proof is $C = 56,116,800$.

For $\tau_0 = 0$, this is the source theorem (up to an inessential change of its universal constant). Printed p. 7, Remark 13, of the source explicitly predicts (2): remove separability and put $\text{tr}_0(K)/m$ inside the square root on the right-hand side [1, p. 7, Remark 13].

2. Idea of the proof

The compact spectral theorem splits

$$H = H_1 \oplus H_0, \quad H_1 = (\ker T)^\perp,$$

where H_1 is separable and carries all nonzero singular directions, while every $h \in H_0$ is zero in L_2 . It is tempting simply to discard H_0 , but that is wrong: a function which vanishes almost everywhere can still have nonzero values at the finitely many sampling nodes and can corrupt a least squares fit.

We add the normalized diagonal $K_0(x, x)/\tau_0$ as a third component of the source’s leverage-score density, alongside the leading and tail spectral components. This keeps the separable feature-vector norm below $3m$ and ensures

$$\frac{K_0(x, x)}{\rho(x)} \leq 3\tau_0. \quad (3)$$

For independent nodes, the off-diagonal values $K_0(x_i, x_j)$ vanish almost surely. Hence the entire weighted H_0 sampling Gram matrix is diagonal and has norm at most $3\tau_0$. The source’s Kadison–Singer step then reduces an arbitrarily large good random draw to $O(m)$ nodes, while its lower frame bound turns (3) into the desired τ_0/m error. The spectral tail is handled exactly as in the source.

3. A parameterized subsampling lemma

We use the equal-weight Kadison–Singer partition lemma employed in Section 4 of [1]. The following is the same argument with the norm budget changed from $2m$ to $3m$ and with a leading block of dimension $r \leq m$.

Lemma 2. *Let $N > 600m$, $1 \leq r \leq m$, and $y_1, \dots, y_N \in \mathbb{C}^r \oplus \ell_2$ satisfy $\|y_i\|_2^2 \leq 3m$ and*

$$\left\| \frac{1}{N} \sum_{i=1}^N y_i y_i^* - \begin{pmatrix} I_r & 0 \\ 0 & \Lambda \end{pmatrix} \right\| \leq \frac{1}{2}, \quad \|\Lambda\| \leq 1. \quad (4)$$

There is $J \subseteq \{1, \dots, N\}$ with $|J| \leq 64800m$ such that

$$\left(\sum_{i \in J} y_i y_i^* \right)_{<_r} \geq 75mI_r, \quad \sum_{i \in J} y_i y_i^* \leq 32400mI. \quad (5)$$

Proof. Compress the lower coordinates to the span of the N lower parts of the y_i . This makes the problem finite-dimensional without changing the vectors or the leading block. If $E = \text{diag}(I_r, \Lambda)$, split the positive matrix $I - E$ into rank-one terms, repeat each term sufficiently many times, and scale the copies so that the artificial vectors have squared norm at most $\delta := 3m$. They have zero leading coordinates. As in [1, Section 4], after adding them the total frame operator M obeys

$$\frac{N}{2}I \leq M \leq \frac{3N}{2}I.$$

The partition lemma used in the source says that if $\alpha I \leq M \leq \beta I$ and $\alpha > 100\delta$, the vectors can be partitioned into blocks J_s with

$$25\delta I \leq \sum_{i \in J_s} y_i y_i^* \leq 3600 \frac{\beta}{\alpha} \delta I.$$

Here $\alpha = N/2$, $\beta = 3N/2$, and $\delta = 3m$, so every block lies between $75mI$ and $32400mI$. If there are t blocks, the lower bound $M \geq (N/2)I$ and the block upper bounds give $t \geq N/(64800m)$. Some block therefore contains at most $N/t \leq 64800m$ of the original vectors. Retain those original vectors and delete its artificial vectors. Deletion preserves the upper bound, and the leading lower bound is unchanged because all artificial vectors have zero leading coordinates. Undoing the compression proves (5). $\square$

4. Proof of Theorem 1

Proof. If $T = 0$, the zero algorithm has zero L_2 error, so assume $T \neq 0$. Choose an orthonormal basis (ϕ_k) of H_1 consisting of right singular vectors and an orthonormal system (b_k) in L_2 such that

$$\phi_k = \sigma_k b_k$$

as functions. This countable system gives the pointwise RKHS expansion of the separable space H_1 .

Fix a sampling budget parameter $m \geq 1$ and put $r = \min\{m, \text{rank } T\}$. Define the active probability-density components

$$A(x) = \frac{1}{r} \sum_{k < r} |b_k(x)|^2, \quad B(x) = \frac{1}{S} \sum_{k \geq r} \sigma_k^2 |b_k(x)|^2, \quad C_0(x) = \frac{K_0(x, x)}{\tau_0}, \quad (6)$$

where $S = \sum_{k \geq r} \sigma_k^2$, and omit B if $S = 0$ and C_0 if $\tau_0 = 0$. If $q \leq 3$ components remain, let ρ be their average. Each active component integrates to one, so $d\nu = \rho d\mu$ is a probability measure. In particular, $K_0(x, x)/\rho(x) \leq q\tau_0 \leq 3\tau_0$ whenever $\tau_0 > 0$.

If $S > 0$, put

$$\gamma^2 = \max \left\{ \sigma_r^2, \frac{S}{m} \right\}; \quad (7)$$

if $S = 0$, put $\gamma = 0$ and omit all tail coordinates. For $S > 0$ define the separable feature vector

$$y(x) = \rho(x)^{-1/2} \left((b_k(x))_{k < r}, (\gamma^{-1} \sigma_k b_k(x))_{k \geq r} \right). \quad (8)$$

Since $r \leq m$ and $S/\gamma^2 \leq m$, (6) gives

$$\|y(x)\|_2^2 = \frac{rA(x) + (S/\gamma^2)B(x)}{\rho(x)} \leq qm \leq 3m. \quad (9)$$

Orthogonality of the b_k gives

$$\mathbb{E}_\nu[y(X)y(X)^*] = \text{diag}(I_r, (\sigma_k^2/\gamma^2)_{k \geq r}), \quad (10)$$

whose lower block has norm at most one.

The random trace-class operator $y(X)y(X)^*$ has trace norm at most $3m$. The strong law in the separable trace class therefore yields an infinite iid sequence for which the empirical covariance converges in trace norm, hence in operator norm, to (10). Also, for each fixed $z \in D$, $K_0(\cdot, z) \in H_0$ and is zero μ -almost everywhere. Since $\nu \ll \mu$, Tonelli's theorem gives

$$K_0(X_i, X_j) = 0 \quad (i \neq j) \quad (11)$$

almost surely for an iid sequence. Choose one realization satisfying (11) and then $N > 600m$ large enough that (4) holds. Lemma 2 produces a subset J of at most $64800m$ nodes satisfying (5).

Let

$$G = \left(\frac{b_k(x_i)}{\sqrt{\rho(x_i)}} \right)_{i \in J, k < r}, \quad \Phi = \left(\frac{\sigma_k b_k(x_i)}{\sqrt{\rho(x_i)}} \right)_{i \in J, k \geq r}.$$

The two conclusions of (5) imply

$$G^*G \geq 75mI_r, \quad \|G^+\|^2 \leq \frac{1}{75m}, \quad \|\Phi\|^2 \leq 32400m\gamma^2. \quad (12)$$

Let $R_0 : H_0 \rightarrow \mathbb{C}^J$ be weighted evaluation, defined by

$$(R_0 h)_i = \frac{h(x_i)}{\sqrt{\rho(x_i)}}.$$

Its row Gram matrix has (i, j) entry $K_0(x_i, x_j)/\sqrt{\rho(x_i)\rho(x_j)}$. By (3) and (11), it is diagonal and

$$\|R_0\|^2 \leq 3\tau_0. \quad (13)$$

Use the weighted least-squares algorithm

$$\mathcal{A}f = \sum_{k < r} \left(G^+ \left(\frac{f(x_i)}{\sqrt{\rho(x_i)}} \right)_{i \in J} \right)_k b_k. \quad (14)$$

It is linear and exact on $\text{span}\{b_k : k < r\}$. Decompose $f = P_r f + Q_r f + f_0$ according to the leading singular directions, the spectral tail, and H_0 . The residual sampling map on $H_{1, \geq r} \oplus H_0$ is the row operator $[\Phi \ R_0]$, so (12)–(13) give

$$\|[\Phi \ R_0]\|^2 = \|\Phi\Phi^* + R_0 R_0^*\| \leq 32400m\gamma^2 + 3\tau_0. \quad (15)$$

Pythagoras in L_2 , exactness on the leading space, and $\|Q_r f\|_{L_2} \leq \sigma_r \|Q_r f\|_H$ now yield, for $\|f\|_H \leq 1$,

$$\begin{aligned} \|f - \mathcal{A}f\|_{L_2}^2 &\leq \gamma^2 + \frac{32400m\gamma^2 + 3\tau_0}{75m} \\ &= 433\gamma^2 + \frac{\tau_0}{25m}. \end{aligned} \quad (16)$$

The same computation covers finite rank when the tail is absent.

Monotonicity of the singular values gives

$$\gamma^2 \leq \frac{2}{m} \sum_{k \geq \lceil m/2 \rceil} \sigma_k^2. \quad (17)$$

Consequently,

$$g_{64800m}(F)^2 \leq \frac{866}{m} \sum_{k \geq \lceil m/2 \rceil} d_k(F)^2 + \frac{\tau_0}{25m}. \quad (18)$$

Apply (18) with $m = 866n$. Since the tail beginning at $433n$ is no larger than the tail beginning at n , and sampling numbers decrease with the number of nodes,

$$g_{(64800)(866)n}(F)^2 \leq \frac{1}{n} \sum_{k \geq n} d_k(F)^2 + \frac{\tau_0}{n}.$$

As $(64800)(866) = 56,116,800$, this proves the theorem. $\square$

5. Checks, dependence, and scope

- **Null-space necessity.** Although H_0 is invisible in L_2 , its point values can perturb least squares. Bound (13), not deletion of H_0 , is what justifies the extension.
- **Degenerate cases.** If $T = 0$, the zero algorithm is exact in L_2 . If T has finite positive rank, the same proof takes all positive singular directions once m exceeds the rank; (13) still gives the τ_0/m term. If $\tau_0 = 0$, the third density component and (13) vanish.
- **External dependency.** The only deep imported step is the equal-weight partition lemma used by Dolbeault–Krieg–Ullrich, ultimately coming from the Marcus–Spielman–Srivastava solution of Kadison–Singer. The compact spectral decomposition and definition (1) are the framework of Moeller–Ullrich [2, Section 7].
- **Nonconstructive character.** The trace-class strong law and the Kadison–Singer partition supply existence, not an efficient algorithm. The numerical constant is a bookkeeping constant, not a sharp claim.
- **Scope.** This proves only the explicit nonseparable-RKHS extension in Remark 13. It does not address the paper’s broad statement that L_p approximation for $p \neq 2$ is widely open.

6. Provenance and literature search

Moeller–Ullrich already prove a nonseparable estimate with the same $\text{tr}_0(K)$ obstruction, but their Theorem 1.2 retains logarithmic oversampling/tail loss [2, Theorem 1.2 and Section 7]. The present argument combines their null-kernel density and diagonal-Gram observation with the later sharp subsampling mechanism of [1]; neither cited paper states (2).

Exact-title, exact-symbol ($\text{tr}_0(K)$), nonseparable-RKHS, and citing arXiv searches were run on 9 August 2026. They found no later paper stating this theorem. That is evidence against an obvious duplicate, not a certified priority or exhaustive-literature claim.

References

- [1] M. Dolbeault, D. Krieg, and M. Ullrich, *A sharp upper bound for sampling numbers in L_2* , Appl. Comput. Harmon. Anal. 63 (2023), 113–134; arXiv:2204.12621.
- [2] M. Moeller and T. Ullrich, *L_2 -norm sampling discretization and recovery of functions from RKHS with finite trace*, Sampl. Theory Signal Process. Data Anal. 19 (2021), Article 13; arXiv:2009.11940.
- [3] A. W. Marcus, D. A. Spielman, and N. Srivastava, *Interlacing families II: mixed characteristic polynomials and the Kadison–Singer problem*, Ann. of Math. 182 (2015), 327–350.